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Creating Electronic Books with Code and Artificial Intelligence Assistants

Course for the Faculties of Sciences and Chemical Sciences

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Welcome to **Creating Electronic Books with Code and Artificial Intelligence Assistants**.

What is this?

This is the course material and a template designed for the teaching staff of the **Faculties of Sciences and Chemical Sciences at USAL** to create interactive teaching books easily. The main idea of this template is for it to be broad enough to cover many use cases and for each student to adapt it to their own course, while also being sufficiently equipped to be used easily with the help of AI assistants (such as GitHub Copilot, Gemini, Claude, Codex, etc.) and a code editor such as VS Code.

Content

In this book you will find:

- *Tutorials* to learn how to use the template
- *Examples by Degree* to see real cases
- Information on *how to cite* and *licenses*

PDF version

You can also download the printable version of the book:

- Download PDF in English
- Descargar PDF en español

Note

This project is designed to be used with **VS Code** and **AI** assistants.

Part I

Example

SOLVING QUADRATIC EQUATIONS

This example chapter shows how to solve quadratic equations of the form

$$ax^2 + bx + c = 0$$

Assuming $a \neq 0$, the solutions are given by the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Quick example:

- For $x^2 - 3x + 2 = 0$ we have $a = 1, b = -3, c = 2 \rightarrow x = 1$ or $x = 2$.

Short step-by-step explanation and links to resources.

(Traducción al español disponible en la versión en español del libro.)

WHAT IS INTEGRATION BY PARTS?

Integration by parts (IBP) is a technique to compute integrals of the form $\int u \, dv$, based on the product rule for derivatives.
Formula:

$$\int u \, dv = uv - \int v \, du$$

Choose u and dv so that $\int v \, du$ is simpler.

USING THE RESOURCES

In the repository `recursos` folder there are several PDFs used as reference:

- `recursos/IBP.pdf` — theory and basic examples.
- `recursos/Kira2.pdf`, `recursos/Kira3.pdf` — `kira` manuals.
- `recursos/FIRE5.pdf`, `recursos/FIRE6.pdf` — `fire` manuals.

Below are practical examples showing how to use `kira` and `fire` to assist IBP tasks (integral reduction, symbolic manipulation, simplification).

EXAMPLE 1 — A STEP-BY-STEP INTEGRAL

Compute $\int xe^x dx$.

1. Let $u = x$, $dv = e^x dx$. Then $du = dx$, $v = e^x$.
2. Apply the formula: $\int xe^x dx = xe^x - \int e^x dx = xe^x - e^x + C$.

EXAMPLE 2 — USING KIRA FOR SYMBOLIC MANIPULATION

Assuming `kira` can read simple expressions and apply IBP automatically:

```
# Apply IBP automatically and save result
kira --input "integrate(x*exp(x), x)" --method ibp --output result.txt
cat result.txt
```

Expected output:

```
x*e^x - e^x + C
```

Consult `recursos/Kira2.pdf` and `recursos/Kira3.pdf` for detailed options.

EXAMPLE 3 — USING FIRE TO REDUCE INTEGRAL FAMILIES

`fire` can reduce parameter-dependent integrals and find relations between them.

```
# Reduce family  $I(a) = \int x^a e^x dx$ 
fire --integral "I(a) = integrate(x^a * exp(x), x)" --reduce --param a --output fire_
  ↪ result.txt
cat fire_result.txt
```

See `recursos/FIRE5.pdf` and `recursos/FIRE6.pdf` for full syntax and advanced examples.

REPRODUCIBILITY NOTES

- Reference PDFs are in `recursos/` and are not committed by default. Ensure readers have them locally.
 - If desired, we can extract small licensed excerpts into Markdown with proper attribution.
-

Part II

Information

ABOUT

You can cite this book as:

Álvaro Lozano Murciego and André Sales Mendes (2025). *Creating Electronic Books with Code and Artificial Intelligence Assistants*. https://elloza.com/teachbook_usal_template/ . Source files at https://github.com/elloza/teachbook_usal_template . CC BY 4.0.

You can also cite individual pages of the book as:

<Page title>. In Álvaro Lozano Murciego and André Sales Mendes (2025). *Creating Electronic Books with Code and Artificial Intelligence Assistants*. https://elloza.com/teachbook_usal_template/ . Source files at https://github.com/elloza/teachbook_usal_template . CC BY 4.0.

Because this book is expected to evolve over time, it is recommended to cite the source repository as well whenever the exact context of the material matters.

8.1 How this book is made

This website is written in Markdown files and Jupyter notebooks, and converted to HTML using tools from [TeachBooks](#) and [Jupyter Book](#).

The source files are stored in a public GitHub repository:

- https://github.com/elloza/teachbook_usal_template

The published version of the book can be viewed at:

- https://elloza.com/teachbook_usal_template/

8.2 About the authors

8.2.1 Authors

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8.2.2 Institution

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CREDITS AND LICENSES

9.1 Project credits

This book and the course template are built with [TeachBooks](#), on top of the [Jupyter Book](#) and [MyST Markdown](#) ecosystem.

TeachBooks provides documentation, components and working patterns that make it easier to create open, interactive and reproducible teaching books.

9.2 Acknowledgements

We thank the [TeachBooks](#) project and [Delft University of Technology \(TU Delft\)](#) for their work on open tools for educational books. The [credits page in the TeachBooks Manual](#) is used as a reference for explicitly acknowledging the tools, communities and institutional support on which this material builds.

9.3 Book license

This book is distributed under the [Creative Commons Attribution 4.0 International \(CC BY 4.0\)](#) license.

This allows the material to be shared and adapted, as long as attribution is provided.

9.4 External resources

Unless explicitly stated on a specific page, the content of this book is authored by the project authors.

If any external resources are reused in a page, they should be acknowledged on that page together with their corresponding attribution and license.

9.5 Code and tools

This project uses external tools and libraries, including:

- [TeachBooks](#)
- [Jupyter Book](#)
- [MyST Markdown](#)

Each tool keeps its own license.

HOW TO CITE

If you use this material, please cite it as follows:

10.1 Text Citation

Faculties of Sciences and Chemical Sciences (2026). *Creating Electronic Books with Code and Artificial Intelligence Assistants*. University of Salamanca. Available at: https://github.com/elloza/teachbook_usal_template

10.2 BibTeX

```
@book{creating_electronic_books_ai_2026,  
  author = {Faculties of Sciences and Chemical Sciences},  
  title = {Creating Electronic Books with Code and Artificial Intelligence Assistants}  
  ↩,  
  year = {2026},  
  publisher = {University of Salamanca},  
  url = {https://github.com/elloza/teachbook_usal_template}  
}
```

10.3 DOI

(Pending assignment via Zenodo)

BIBLIOGRAPHY